THIRD UNCERTAINLP WORKSHOP · EMNLP 2026 · BUDAPEST

Semantic Calibration Prevails Where Token Confidence Fails

Benchmarking Long-Form Scientific QA

Philip Müller[1] **Nicholas Popovič**[2,3] **Michael Färber**[2,3] **Peter Steinbach**[1]

[1]Helmholtz-Zentrum Dresden-Rossendorf [2]Technische Universität Dresden [3]ScaDS.AI Dresden/Leipzig

We benchmark the calibration of four uncertainty quantification methods on 685,000 long-form answers from up to 20 LLMs. Instruction tuning polarizes token probabilities: nearly all answers receive the same confidence, a collapse that ECE can hide. Reasoning models diverge by provider. Only semantic consistency of the final answer stays calibrated in multi-step reasoning.

Reasoning Demands

7 datasets · 3 tiers

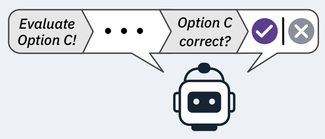
Recall
MMLU · SciQ · ARC-Easy

Multi-step
ARC-Challenge · GSM8K · SVAMP

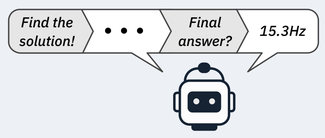
High complexity
SciBench · GPQA

Task Formats

Multiple choice
each option judged separately by CoT + counterfactual prompting (APriCoT)



Arithmetic
chain of thought, final number extracted



Long-Form Generation

685,000 responses

Up to 20 open-weight LLMs
5 providers · base, instruct, reasoning
7B–70B · dense & MoE

mean answer length 127–6,827 tokens, depending on dataset and model

10 samples per prompt

generated once and cached: all methods score **identical outputs**

Uncertainty Quantification

calibration, not ranking

Experiment 1 · token level

Label probability probability of the chosen label

Experiment 2 · sequence level

Verbalized stated confidence
P(True) p("True") for its own answer
CCP NLI-weighted token product
Answer Consistency agreement across 10 samples

Token level

RQ1 Are label probabilities calibrated, and how do instruction tuning and reasoning change them?

1 Instruction tuning polarizes token probabilities

Illustration with synthetic data

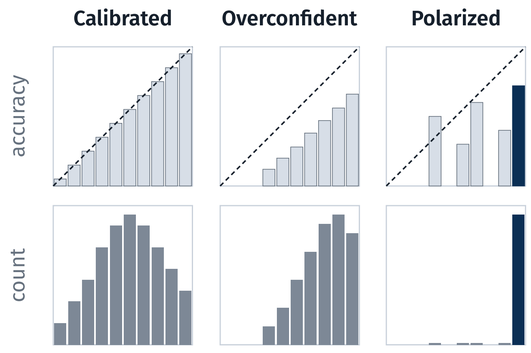


Figure 1. Synthetic illustration: accuracy (top) and answer count (bottom) per confidence bin.

Overconfident

Scores exceed accuracy but still rank answers correctly. Platt scaling or temperature scaling is a viable mitigation.

Polarized

The majority of answers receive a similar score. The ranking is lost, and no (re-)calibration can recover it.

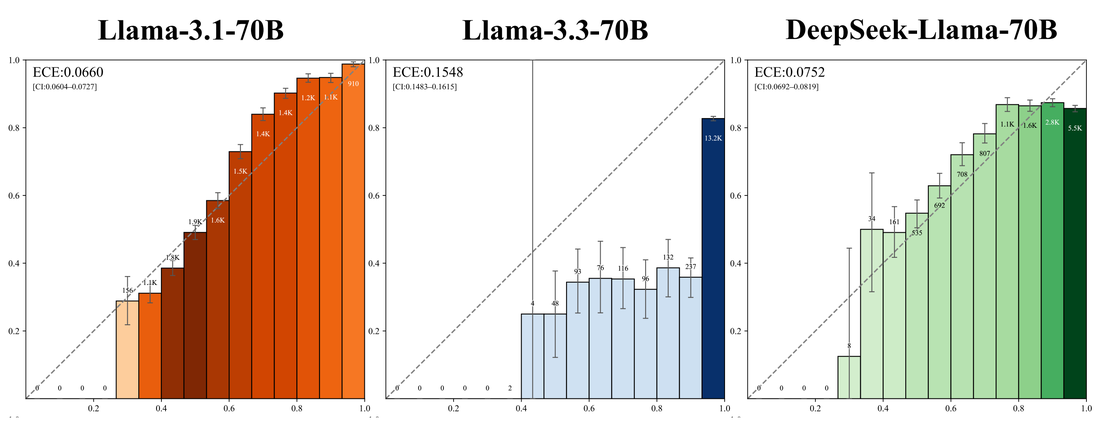


Figure 2. Label-probability calibration on MMLU for base, instruct and reasoning variants of Llama-3 (70B). The instruct model places `13.2k` answers in the top bin; the DeepSeek-R1 distill spreads them out again. Seen across five model families and four datasets; Mistral is the exception.

2 A low ECE does not mean calibrated

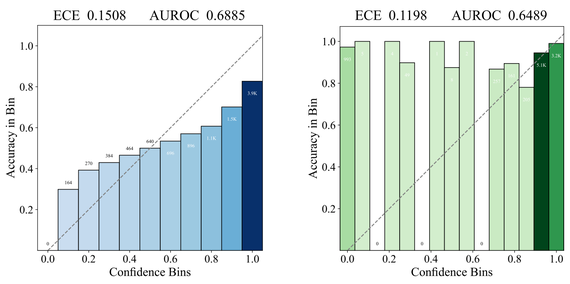


Figure 3. Informative scores (left, ECE 0.151) vs scores unrelated to correctness (right, ECE 0.120).

Once scores collapse, ECE tracks **overall accuracy**, not whether individual scores are informative.

The uninformative model on the right gets the **better** ECE.

Report calibration plots and score distributions alongside ECE.

3 Reasoning models split by provider

Commit early
`gpt-oss` · `Magistral`
Settle on one option early; probabilities **polarize** as in instruct models.

Weigh alternatives
`DeepSeek-R1` · `Qwen3-Thinking`
Weigh several options; probabilities **stay spread**, but traces get longer.

Split follows provider, not scale. Observational: training pipelines are undisclosed.

Sequence level

RQ2 Do sequence-level methods give reliable confidence for long-form answers?

4 Only Answer Consistency is calibrated

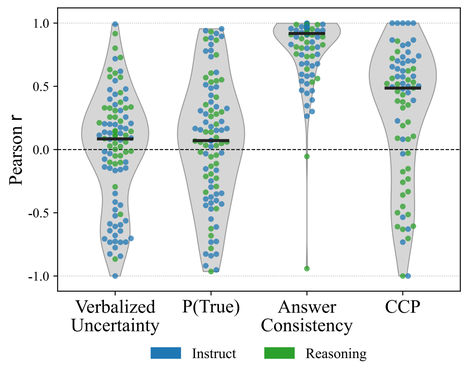


Figure 4. Pearson *r* between bin confidence and bin accuracy; one dot per (model, dataset) pair.

Table 1. Median Pearson correlation coefficient *r* of calibration.

Method	Paradigm	median *r*
Verbalized	self-evaluation	0.08
P(True)	self-evaluation	0.07
CCP†	token aggregation	0.49
Answer Consistency	semantic consistency	**0.92**

†Based on few non-empty bins, as almost all CCP scores are near 0.

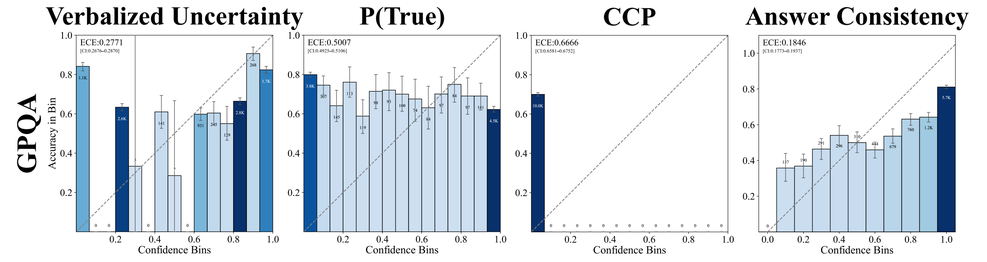


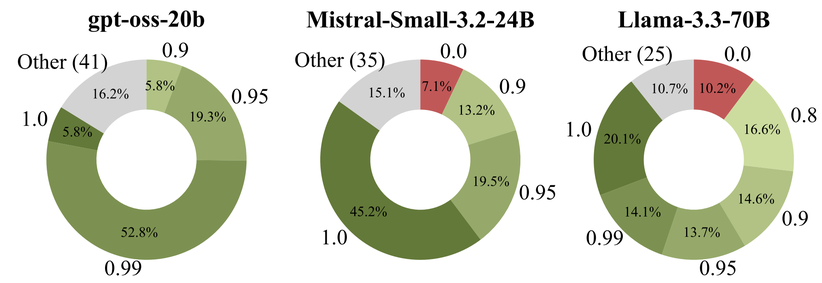
Figure 5. Calibration of the four methods for `Llama-3.3-70B-Instruct` on GPQA. Only Answer Consistency follows the diagonal.

5 Why single-trajectory signals fail

Table 2. Failure modes of the sequence-level methods.

Method	Behavior on long-form answers
Verbalized	Uses only a few fixed values, regardless of correctness.
P(True)	Relies on token probabilities; inherits polarization if present (1).
CCP	A product over thousands of tokens drives every score toward 0.
Answer Consistency	Agreement across 10 samples. Caveats: shared errors, ambiguous questions, 10× cost.

(a) Verbalized confidence values



(b) P(True) score histograms

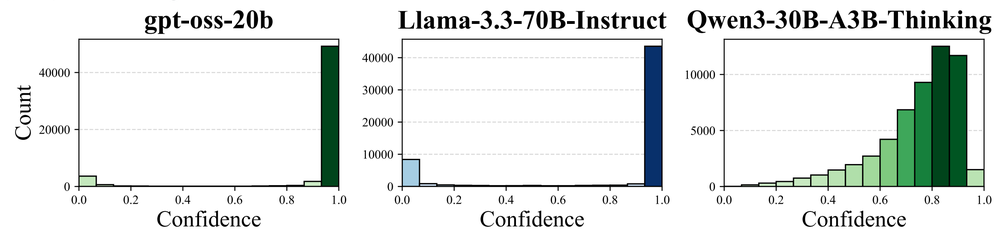


Figure 6. Confidence scores over all datasets. (a) A few values near 1.0 dominate. (b) Scores concentrate at 1.0, except for `Qwen3-Thinking`.

For practitioners

1. Check token-level scores for polarization before thresholding on them.
2. Report calibration plots and score distributions, not ECE alone.
3. Need calibrated confidence? Sample several answers and measure their agreement.

Limitations

- Structured tasks with verifiable answers; English only.
- Observational: no ablations of training stages.
- Fixed decoding; answer clustering must be tractable.

Paper & contact

`arxiv.org/abs/2602.00279`
`philipmueller.dev@gmail.com`
`philipmueller.dev`

Supported by Helmholtz AI and the participating institutions, which also provided computing resources.